

An Entanglement Perspective on How Older Adults Use and Design Emerging Ubiquitous Technologies

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ABSTRACT

Over the last decade, CSCW and HCI researchers have adopted different user-centered approaches to include older adults' perspectives in designing the next generation of emerging technologies such as virtual reality, Internet of Things technologies, and conversational voice-based AI assistants. However, a body of emerging research is showing what older adults share in such user-centered design activities might not reflect their actual needs or preference. Hence, for my dissertation I seek to go beyond what older adults simply say or do in user-centered design activities and understand the larger socio-material configurations in which older individuals and their technologies participate. To do this, I use the Entanglement HCI framework to analyze data from interviews and co-design activities with older adults focused on the topic of use and design of conversational AI assistants. In doing so, my dissertation will provide an empirical understanding of a) the different actors (human and non-human) that configure older adults' use and perception of emerging technologies, b) identify implications for designing conversation voice technologies for older adults, and c) provide a preliminary understanding of what Entanglement HCI has to offer for understanding some challenging phenomena of Aging in CSCW and HCI, and thereby provide new perspectives on how we could include older adults in designing the next generation of technologies.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **User characteristics**; • **Social and professional topics**; • **Seniors**;

KEYWORDS

Entanglement HCI, older adults, aging, fourth wave, ubiquitous technologies, conversational voice assistants

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1 INTRODUCTION AND BACKGROUND

With the proliferation of emerging technologies such as virtual reality (VR), augmented reality (AR), or ubiquitous IoT including conversation voice technologies (e.g., Alexa, Google assistant), researchers are considering older adults' preferences and needs with respect to these technologies (e.g., [1, 2, 12, 30]). In this thread of work, much of the research focuses on how these emerging technologies can support older individuals' health and wellbeing (e.g., [15, 30]) or support social roles and interactions (e.g., [6, 19]). Often, a common theme of investigation is to understand if older adults find these technologies useful and how might we design these technologies to better meet older adults' needs and preferences, so they are more likely to find these technologies relevant and adopt them. Yet, research has often shown that older adults might not always find these technologies meaningful [20] and might not be likely to adopt them, at times even considering some of these emerging technologies a "toy" [28]. So, as a way to design technologies that older adults find meaningful, researchers in the CSCW and HCI community have adopted user centered approaches to include older adults in the design process through different methods such as participatory design (e.g., [29]), focus groups (e.g., [28]), interviews (e.g., [4]).

Yet, even with these approaches of including older adults in design, recent research is indicating a dissonance in what older individuals say or express in these activities and what they might actually want. For example, in a five year co-design study, the same older adults who designed a technology did not seem keen on adopting them when they were built [27]. Lazar et al.'s [14] work analyzed older adults' participation in designing ubiquitous smart home technologies through a diffractive lens [11] and some of their findings indicate how older adults might not always be interested in such activities or participate just enough "to satisfy researchers". In my other works, I have found how older adults might think of "other" stereotypical older adults and not themselves in such design activities [23], which implies that even listening to participants might not be enough to evade stereotypes of aging. Collectively, this body of work suggests what older adults express in design activities might not be their actual preference. Given these differences between what people say and what they actually want or might use, it is not entirely straightforward for researchers to understand how older adults perceive, use or want these emerging technologies to be designed.

In this context, Entanglement HCI (or what some researchers are proposing as a Fourth wave in HCI [7, 9, 10]) is a research paradigm that may be able to help us understand phenomena that are challenging to interpret in existing paradigms [7]. This paradigm provides us a lens to go beyond what people simply say or do, to probe and understand the larger socio-material configurations in

which these technologies and humans intra-act, along with other social and material actors. Entanglement HCI argues that with increasing ontological uncertainties posed by emerging technologies (such as conversational AI, IoT, VR, AR), the popular CSCW and HCI concepts of situatedness, context, and embodiment might fail to capture the complex relationships that humans and their technologies form. By decentering the human as the sole source of attention, Entanglement HCI elevates the role of the non-human world and acknowledges non-human actors (e.g., materials, norms, legal rules) as equal contributors to any action (such as use or non-use of technology) as it unfolds. According to Entanglement theories, participation of both human and non-human actors enacts realities and is shaping who we are in this society. This acknowledgement of the participation of non-human actors, in addition to human actors, provides a systematic way of tracing distribution of agency, along with accountability and responsibility.

By decentering the human, the fourth wave lens goes beyond thinking about “user” experience to designing meaningful socio-material configurations in which human and non-human actors (including technology) intra-act. Therefore, it could provide us a nuanced lens to think and understand the different actors that actively shape older adults’ use and perception of emerging technologies. This could guide us towards a future where we can meaningfully include older individuals in designing socio-material configurations in which these emerging technologies intra-act. Motivated by the potential of this new paradigm, below I describe my research objectives.

2 RESEARCH QUESTIONS

In my work, first, I seek to understand how older adults use, perceive, and design ubiquitous technologies. Second, I also seek to investigate how might we better include older adults in design. Ubiquitous technologies spans across a vast arena of emerging technologies (e.g., VR, AR, implants, IoT). For the purpose of my dissertation, I scoped my studies to focus on ubiquitous conversational voice technologies at home. I specifically chose smart speaker-based voice assistants at home such as Alexa on Amazon Echo or Google Assistant on Google Home. This decision was guided by recent works which suggest, in addition to perceived accessibility benefits [24, 25], these conversational voice assistants can play informal social roles as a conversational partners [21, 30]. This suggests how there might be an intimate entanglement between older adults and these conversational assistants, indicating how ubiquitous voice technologies are a form of emerging technology that can benefit from Entanglement HCI thinking.

My work is inspired by the Entanglement HCI thinking that we “don’t really create technologies that functionally serves the needs of people but rather by configuring different material conditions we design humanity” [7]. Hence, going beyond past framings of user-centered approaches that solely considers the user, my work seeks to understand the nuanced socio-material configurations where older adults along with other non-human actors (including voice assistants) intra-act, and how these intra-actions shape individuals’ social relationships, their perception, use, and the design of these technologies. Entanglement HCI also argues that ontological uncertainties are posed by emerging technologies [7]. Hence, I also

seek to understand why and how ontological uncertainties unfold as older adults think of ubiquitous voice technologies.

As such, in my dissertation, taking an Entanglement perspective to analyze data from interviews and co-design activities with older adults, I seek to answer the following questions:

RQ1: How do older adults perceive and ontologically categorize ubiquitous voice assistants?

RQ2: What human and non-human actors participate in socio-material configurations involving older adults and their voice assistants? How do these actors intra-act to shape different realities including social relationships?

RQ3: In co-design activities focusing on designing such ubiquitous technologies, what are the different human and non-human actors that participate? How can we create spaces that enable older adults (and other human and non-human actors) to come together in participatory speculation of future socio-technical configurations that includes ubiquitous voice technologies?

3 WORK IN PROGRESS

Below I describe my progress to date and approach to answer the above research questions. For the first two parts of my dissertation, I focus on understanding the different configurations and the actors that play a role in older adults’ *use and perception* of conversational voice assistants. For my final study, I shift focus to explore how older adults would *design* these technologies and investigating performative roles of both human and non-human actors in such design sessions.

3.1 Older adults’ ontological categorization of voice assistants

Although Entanglement theories argue that ontological uncertainty is posed by emerging technologies [7], it is unclear how this ontological uncertainty unfolds as people use these technologies. With respect to voice assistants, some prior work indicates that people personify these smart speaker-based voice assistants by describing them using personal pronouns [17, 26], but little is known about how people ontologically categorize these assistants, i.e., do they think of these technologies as “human-like” or “object-like”. To understand how older adults perceive and ontologically categorize ubiquitous voice technologies (RQ1), I analyzed interview and diary study transcripts from a three-week deployment study of voice assistants (i.e., Amazon Echo Dot) in older adults’ homes. The detailed methods and findings can be found here, published at CSCW [22].

From this work, I learned that ontological uncertainty unfolds when older adults use these ubiquitous voice assistants. We found that instead of straightforwardly categorizing the voice assistant as “human-like” or “object-like”, participants had a nuanced categorization where they fluidly moved between the two ontological categories. The findings of this work also uncover different human and non-human actors playing a role in this ontological movement, including: the characteristics of the voice assistant (such as nature of interaction), the user themselves (desire for social contact or affiliation), duration of using the voice assistant, location of the voice assistant in the house and the perceived presence of the voice assistant.

3.2 Actors participating to enact different realities and phenomena

In my second study, I seek to investigate another key tenet of Entanglement HCI – the participation of non-human actors in addition to human actors in enacting certain phenomena [7]. In the paper Entanglement HCI, Frauenberger argues that we as technologists are not designing technologies, nor can we design interactions. Rather, according to the Entanglement HCI “what we seem to be doing is *creating configurations* that enact certain phenomena and realities” [7]. Here configurations are hybrid networks where different human and non-human actors intra-act [7, 13]. Hence, it is important to understand the different socio-material configurations in which humans and their technologies (along with other actors) intra-act to enact different phenomena (which at times may be desired or undesired).

Therefore, in the second phase of my work, I seek to understand the different human and non-human actors that intra-act to enact different phenomena in socio-material configurations involving older adults and their voice assistants (RQ2). Towards this, I have conducted semi-structured interviews with sixteen older adults who own and use a smart speaker-based voice assistant to understand the current usage of their voice assistant. Currently, I am analyzing the data through the Entanglement perspective to investigate the following: a) when and how different human or non-human actors participate in enacting phenomena involving older adults and their voice assistants, b) how voice technologies at home configure older individuals’ social lives, and c) how older adults (re)-configure voice assistant use. The findings of this work will provide a preliminary understanding of how older adults form relationships with voice assistant technologies, the different ways in which older adults’ and voice assistant technologies constantly re-configure each other. This work will also help identify the different human and non-human actors that participate in enacting desired and undesired realities.

3.3 Including older adults in participatory speculation of future socio-technical configurations

With the intimate entanglement between humans and their technologies, Frauenberger recommends going beyond user-centered design to develop participatory speculation approaches with users to design meaningful relations, rather than “optimizing user experiences” [7]. Suggested approaches to do this is to evolve our traditional participatory design method. Some of the ways to do this include identifying the participation of different human and non-human actors, tracing associations between them, and paying close attention to controversies as they emerge [5, 7].

With this in mind, for my final dissertation study, I seek to identify the different human and non-human actors that participate in co-design activities involving older adults in designing ubiquitous technologies (RQ3). As a part of this, I also seek to understand approaches to create spaces that can enable older adults (and other actors) to come together in participatory speculation of socio-technical configurations. To answer my third research question, I will conduct a series of co-design workshops with older adults on designing conversational AI assistants. I chose this approach

because co-design workshops have been a popular approach to include older adults in design of future technologies [3, 16, 29]. These workshops will be focused around designing the fictional character for the voice assistant since it acts as a guide to write conversations for the assistant [8], and/or writing conversation dialogues for the voice assistant (to understand how older adults would want the agent to talk in their homes).

As older adults engage in designing the AI assistants, I will pay close attention to the controversies that emerge in the activities and identify the different socio-material actors that participate in these design sessions. Additionally, I will also analyze the performative role of different materials we use in our design activities, (e.g., cards, whiteboard, paper journal) and how these materials can support or restrict actions towards creating a space where older adults can speculate future socio-technical configurations.

4 EXPECTED CONTRIBUTIONS

My dissertation research will use Entanglement perspectives [7] to understand the nuanced socio-material configurations where older adults along with other non-human actors (including voice assistants) intra-act, and how these intra-actions shape individuals’ social relationships, perception, use, and design of these technologies. The expected contributions of this work are:

- a) Provide an empirical understanding of the different human and non-human actors that configure older adults’ use and perception of voice assistants. By going beyond what older adults simply say or do in user-centered activities, this work will help provide a nuanced understanding of how different actors configure individuals’ use and perception of these technologies.
- b) By taking into consideration the participation of non-human actors, in addition to human actors, this work will help identify new implications for designing conversational voice technologies for older adults.
- c) By including older adults in co-designing emerging technologies, and analyzing these interactions through an Entanglement lens [7], this work will contribute to the growing literature on including older adults in design through co-design and participatory approaches (e.g., [16, 18]).
- d) Provide a preliminary understanding of what Entanglement HCI has to offer for understanding Aging CSCW and HCI, especially as we begin designing the next generation of ubiquitous technologies.

5 DOCTORAL CONSORTIUM PARTICIPATION

As a participant of the CSCW doctoral consortium, I would greatly benefit from the feedback of experienced researchers and peers with socio-technical backgrounds. CSCW researchers have often used qualitative approaches to critically analyze data using theoretical frameworks, with the aim of better understanding the social, material, and theoretical challenges of designing technology. My work aligns with this thread of CSCW, where I am engaging with older adults, a user group who have traditionally been underrepresented in technology design. Hence, the feedback I receive from CSCW researchers would be valuable for my work. As a participant,

I would also engage in providing constructive feedback to my peers. By the time the consortium takes place, I would be finalizing my final study on the co-design workshops with older adults. Therefore, it would be an idea time for me to get feedback from researchers. I would value feedback regarding my methodology, study design, possible shortcomings or limitations of my approach.

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